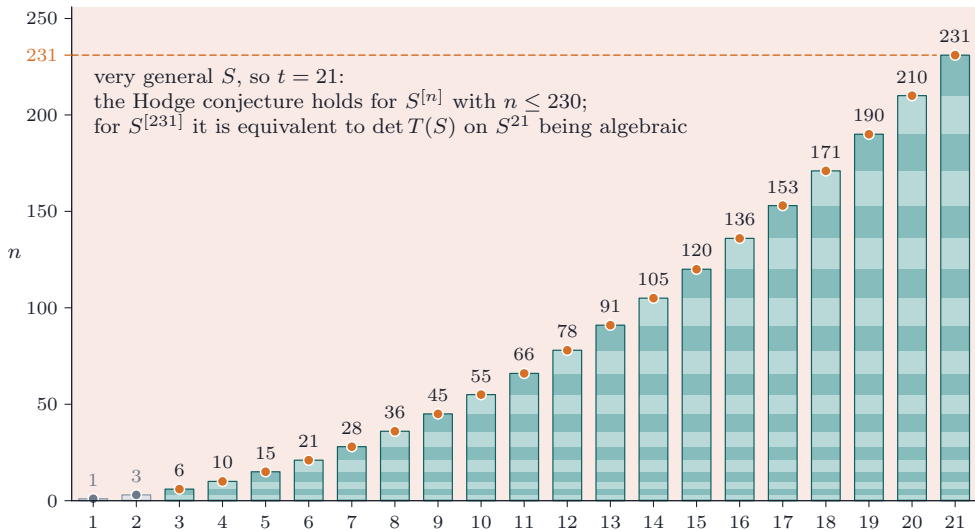




t : the number of distinct part sizes of a partition of n ; for S with $E = \mathbb{Q}$, $t = \dim T(S)$

- $n < t(t+1)/2$: the Hodge conjecture holds for $S^{[n]}$; bar t is cut into the parts $1, 2, \dots, t$
- $n = t(t+1)/2$: the Hodge conjecture for $S^{[n]} \Leftrightarrow \det T(S)$ on S^t is algebraic $\Leftrightarrow$ it holds for every S^k
- $n > t(t+1)/2$: open; it follows from the algebraicity of $\det T(S)$ on S^t
- $t \leq 2$ does not occur for a K3 surface with $E = \mathbb{Q}$